

Text Steganography on Sundanese Script using Improved Line Shift Coding

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Abstract— Steganography allows confidential information to be inserted through multimedia (text, video, image and audio). Text steganography changes the text format or the character of a particular text without changing the meaning of the text. Sundanese script, which already registered in the official version of the Unicode font, can be used as an alternative media to be a cover text in text steganography technique. Text steganography technique has the vulnerability to experience information loss due to attack or malfunction such as printing process with the printer and reprinting by copy machine. The line shift coding approach is used to produce good robustness level stego text and reduce the damage caused by the printing and scanning process. Line shift coding inserts binary bits of secret messages by vertically shifting the row position. The insertion of confidential information in Sundanese script steganography text replaces the Unicode character block on the shifted row. Contribution of this research is improving the secret message capacity. In ordinary line shift coding, odd row is the pivot and the secret bit is inserted on the line between those two pivots. In the improved line shift coding, we use the first and the fifth row as the pivot line to increase the message capacity. The experimental results shows the improved line shift coding has higher capacity and the stego text is able to withstand printing process and copying process up to two reprints to retain the confidential messages.

Keywords— *Line Shift Coding, Text Steganography, Sundanese Script.*

I. INTRODUCTION

Information security is an important factor in information and communication technology, where the confidentiality of information needs to be maintained (including the sending information process). Steganography becomes one of the alternative forms of information concealment. The word steganography itself comes from the Greek "stegos" which means the cover and "grafia" meaning writing [1]. Steganography inserts information through media such as text, video, images and audio.

Text steganography is the most difficult type because of the low level of information redundancy compared to audio and image media [2]. Text steganography changing the text format or changing certain text characters without changing the meaning of the text [3]. Vidhya and Vargese use Malayalam for cover text as an alternative method for text steganography [2]. Alla and Prasad use Telugu as a method of text steganography by utilizing Ottulu and punctuation appropriateness [4]. Sundanese script can be used as an alternative cover text in the Indonesian region. Sundanese script has been registered in the official version of Unicode font and can be used as a cover text on steganography process.

The result of steganography technique is assessed with these parameters: fidelity, recovery (the retrieval of confidential information from stego text), and robustness (the stego text resistance level in retaining information from steganalysis) [5]. Steganalysis is used by testing the robustness: photocopy, scaling, or cropping on stego text document [6].

Line shift coding used to reduce the extent of damage to text steganography from steganalysis using printing and scanning [7]. Line Shift Coding is a steganography method by shifting the line vertically to insert secret information in the cover text. Contribution of this research is improving the secret message capacity by using the first and the fifth row as the pivot line in line shift coding algorithm.

This paper is organized as follow: first section is the motivation of the paper, second section is literature study, third section contains improved line shift coding, fourth section is the functional and performance evaluation of the improved line shift coding, fifth section is the conclusion of the paper.

II. LITERATURE STUDY

A. Text Steganography

Steganography is a technique of information concealment where not only the inserted information is confidential, the existence of the hidden information itself has also confidential. Steganography can be the represents as text, images, audio, and video files. In the steganography process, the cover text is a media (container) for confidential information. Embedded text (secret message) is the confidential information which inserted in the cover text. Stego text is the output of the information embedding process. Stego key is a term used to refer to secret keys used to hide confidential information or to extract confidential information from cover text [8].

Text steganography changes the text format or certain characteristics of the textual elements. Text steganography can be classified into three types: Format Based Method, Random and Statistical Generation Method; and Linguistic Method. Format Based Methods changes the physical formatting of text formats to hide information, but are easier to detect. Random and Statistical Generation will hide information randomly by taking into account the statistical properties of characters that appear in text. Linguistic methods specifically consider the linguistic nature of the resulting stego text and modified text cover, and in many cases, use linguistic structures as space to conceal information [9] [10].

B. Sundanese Script

Sundanese script is orthography system of West Java, including character and writing system to write Sundanese language [11]. The Sunda character consists of [12]:

- Independent vowels (a, é, i, o, u, e, and eu) as in Fig 1
- Ngalagena or consonant (ka-ga-nga, ca-ja-nya, ta-da-na, pa-ba-ma, ya-ra-la, wasa-ha, fa-va-qa-xa -za) as in Fig 2
- Number character as in Fig 3
- Rarangkén as a sign of vocalization as in Fig 4

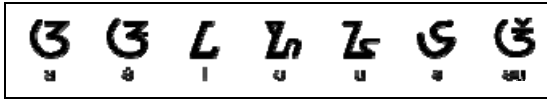


Fig. 1. Example of Sundanese Vowel



Fig. 2. Example of Sundanese Consonant

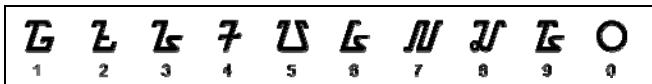


Fig. 3. Example of Sundanese Number

	panyuku, membuat vokal aksara Ngalagena dari [a] menjadi [u]. Contoh: 77 = ka → = ku.
	panyakra, menambah konsonan [r] di tengah suku kata. Contoh: 77 = ka → = kra.
	panyiku, menambah konsonan [l] di akhir suku kata. Contoh: 77 = ka → = kla.

Fig. 4. Example of Rarangken

The Sundanese script has been listed in the official version of Unicode Sundanese Unicode 2013. In this research, Sundanese Unicode font 2013 2nd version is used as the cover text.

C. Line Shift Coding

Line shift coding is a method to change a document by vertically moving the location of a line of text to encode a document. This encoding can be applied either into the file format or to the image bitmap. In certain cases the decoding

process can be done without the original image, since the original image is known to have a uniform line distance between the adjacent lines in paragraph. In line shift coding, only a few lines are shifted, other lines will remain still and become control groups (we can call it pivot line). Line shift coding is able to withstand attacks such as distortion due to scanning with scanners or reprints by copy machines. However, line shift coding has disadvantage in the information capacity [3].

Andiniarti uses the odd line as the control groups in the insertion of information with the line shift coding approach so that even even rows can be moved for insertion of information [7]. To increase the capacity ratio of the line shift coding, we use the first and the fifth row as the pivot line and the secret bits is inserted on the lines between those two pivots.

III. METHOD

A. System Architecture

This system has two modules: embedding and extraction module. Embedding module inserts confidential information into the cover text and generate stego text. Extraction module retrieves confidential information by processing stego text files.

During the embedding module insertion process, the information or secret message will initially be converted into ASCII values and then converted into binary bit form. The length of the converted binary bit will be matched first with the number of shifted rows calculated from the cover text. If the binary bit length is greater than the number of sliding rows, the system will reject insertion. If the binary bit length is less than the number of rows it will add a random binary bit until the binary bit length is equal to the number of sliding rows. The number of shifted rows is affected by the selected shift option. In Even-Line shift options only the even rows that lie between the odd lines are shifted as shown in Fig 5. In the First and Fifth Line Shift Option, the first and the fifth line will be the control groups, the rows between the control groups are shifted as in Fig 6.

The information insertion process shifts the line position and creates PDF stego text document. Before writing the contents of the line into the stego text, rangkén position is normalized to justify the location of rangkén script and correct the rangkén position and shape. Normalization is done by swapping the Unicode blocks of the old rangkén with the new Unicode blocks representing the correct rangkén position.

	1 st Line Fixed
	2 nd Line Shifted
	3 rd Line Fixed
	4 th Line Shifted
	5 th Line Fixed
	6 th Line Fixed

Fig. 5. Line Shift Coding – Even Line Shift Option

ପ୍ରଥମ ଲିନିଆର ଟେକ୍ସଟ୍	1 st Line Fixed
ଦ୍ୱିତୀୟ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	2 nd Line Shifted
ତୃତୀୟ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	3 rd Line Shifted
ଚତୁର୍ଥ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	4 th Line Shifted
ପଞ୍ଚମ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	5 th Line Fixed
ଷଷ୍ଠ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	6 th Line Fixed
ସପ୍ତମ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	7 th Line Fixed
ଅଷ୍ଟମ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	8 th Line Fixed
ନବମ ଲିନିଆର ଟେକ୍ସଟ୍ ଓ ଟେକ୍ସଟ୍ ଟାଏଲ୍ ଟେକ୍ସଟ୍	9 th Line Fixed

Fig. 6. Line Shift Coding – First and Fifth Line Shift Option

Characters on the shifted row are replaced with characters in the new Unicode block to produce vertically shifted row positions. If the secret message binary bit is "1", then the line position will shift up, else the line position will be shifted down. The process of inserting confidential information can be seen in Fig 7.

In the extraction module, the stego text will be read to retrieve information previously inserted in the cover text. The extraction process begins with counting the number of shearlines in the stego text according to the choice of shift options. The extraction process is followed by the conversion of characters in shifted line into Unicode values. The bit-binary will continue to be added according to the number of shifted rows. The system will stop counting binary bits when the number of rows is equal to the binary bit length. The binary bits then converted into an ASCII character to reveal the hidden information. The extraction process can be seen in Fig 8.

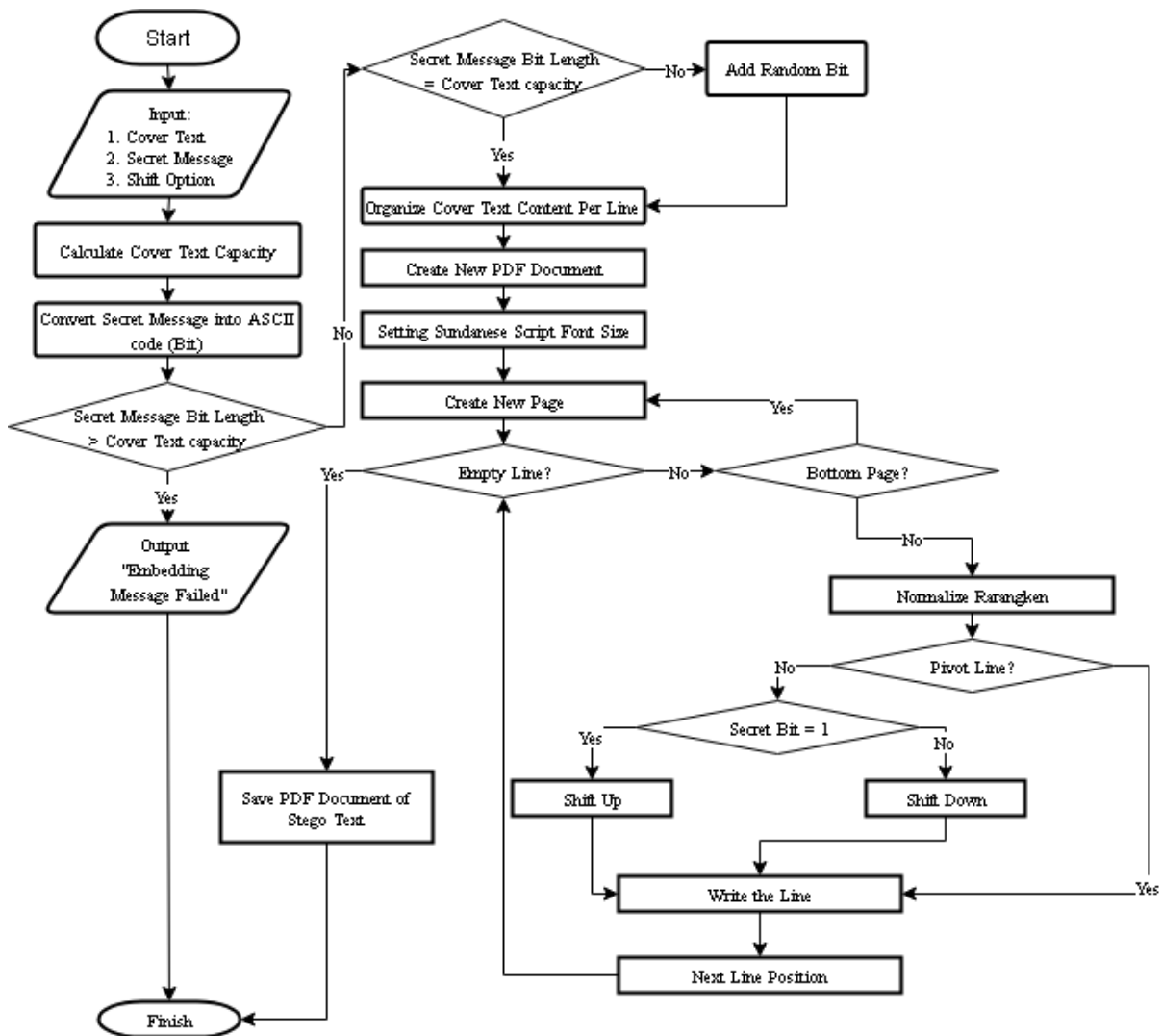


Fig. 7. Embedding Flowchart

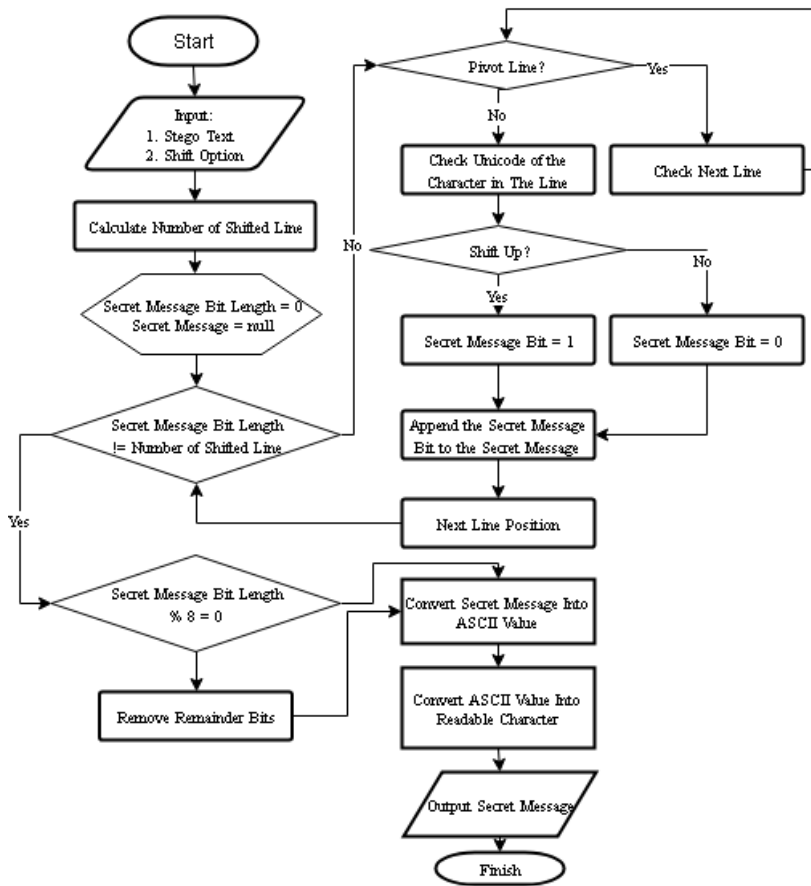


Fig. 8. Extracting Flowchart

B. Application Interface

The system interface is built using JFrame class on Netbeans 8.0.2 in the Java programming language environment. The interface is created for both modules in the system. The embedding module interface can be seen in Fig 9 has the main features for converting confidential information to binary bit form, counting the number of sliding rows and insertion of confidential information into the cover text.

The interface display of the extracting module can be seen in Fig 10, where the extraction module has the main feature for re-extracting confidential information from the stego text.

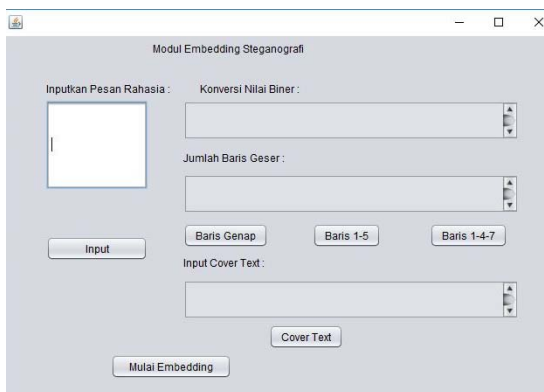


Fig. 9. Embedding Interface

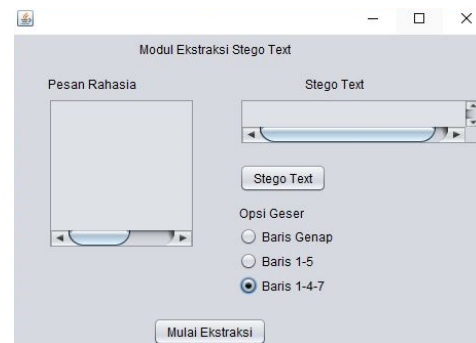


Fig. 10. Extracting Interface

IV. EXPERIMENTAL RESULT AND DISCUSSION

The experiment data can be seen in the Table I. A good steganography produces stego text with a good level of robustness, protects hidden files so they remain unchanged and remain valid, protect the inserted confidential information, stego text as similar as possible with the cover text, and does not cause suspicion. In the extraction process, the resulting stego text will be processed again to search for hidden secret messages. The secret message must be the same as the inserted secret message, so the information is intact and undamaged. Capacity ratio is a measure of media capabilities used as a cover text to accommodate a secret message. Capacity ratio is calculated to measure the ability of Sundanese text in storing the bits of secret message information by referring to the line shift coding approach.

Calculation of capacity ratio (bit / kilobyte) where text capacity (bit) is the capacity of information inserted in the text cover media while text size (kilobytes) is the size of the file inserted information.

TABLE I. EXPERIMENT DATA

No	File Name	File Size (kB)		Source
		Doc	PDF	
1	Anaking.docx	12.7	15.8	[23]
2	Apan Urang Lembur Tea.docx	12.8	16.9	[24]
3	Dandangula Sugan.docx	12.6	16.5	[25]
4	Hidup.docx	12.4	13.9	[26]
5	Katineung.docx	12.3	15.0	[27]
6	Potret.docx	12.5	15.6	[28]
7	Sajak-sajak Godi Suwarna.docx	18.4	25.9	[29]
8	Sekar Merdika.docx	12.5	16.1	[30]
9	Sinareng Bulan Purnama.docx	12.5	16.0	[31]
10	Solawat Daun.docx	12.8	17.3	[32]
11	Sajak Muhsin 500 kb.docx	124.0	531.0	Modified [30]
12	Sajak Muhsin 1mb.docx	243.0	1,044.0	Modified [30]
13	Sajak Muhsin 2mb.docx	473.0	2,095.0	Modified [30]
14	Sajak Muhsin 3mb.docx	704.0	3,128.0	Modified [30]

A. Functionality Experiment

The functionality test on the steganography system is intended to test whether or not the system performs functions such as the conversion of secret messages, the counting of the number of shearlines, the insertion of secret messages, the storage of stego text and the extraction of the secret messages. The result of the functionality test shows that the system succeeds in performing the necessary major functions as in Table II.

TABLE II. FUNCTIONAL EVALUATION

No	Functionality	Result
1	Secret message conversion	Succeed
2	Calculating the number of sliding rows	Succeed
3	Secret Message Insertion	Succeed
4	Stego text storage	Succeed
5	secret message extraction	Succeed

B. Performance Experiment

The performance result of embedding and extracting process can be seen in Table III. From the test results, though not always, the use of even options often has the fastest running time in the embedding and extracting process.

TABLE III. PERFORMANCE EVALUATION

Data	Even Line		1 & 5 Pivot Line	
	Shift Option		Shift Option	
	Embedding (ms)	Extracting (ms)	Embedding (ms)	Extracting (ms)
1	33.49	11.21	33.72	22.65
2	48.79	14.9	32.26	18.42
3	40.87	15.72	49.27	26.07
4	20.88	10	23.56	9.6
5	-	-	-	-
6	-	-	32.9	7.77
7	276.86	32.11	339.77	33.47
8	39.53	6.06	24.01	8.57
9	46.55	11.16	37.9	9.01
10	39.29	12.18	39.78	13.69
11	16,335.43	2,036.74	19,432.78	1,448.09
12	37,521.98	4,241.59	39,854.86	2,837.69
13	109,042.77	7,768.35	143,845.05	6,040.66
14	202,517.35	12,140.06	268,550.01	8,315.38

C. Capacity Ratio Experiment

Capacity ratio testing is calculated by dividing the total number of shear rows with the cover text size. The calculation results are shown in Table IV. The 1&5 pivot line shift option increases the capacity ratio of 1.39 (bits / KB), compared to the use of even-line shift options.

TABLE IV. CAPACITY RATIO

Data	Even Line		1 & 5 Pivot Line	
	Shift Option		Shift Option	
	Capacity (bit)	Capacity Ratio (bit/KB)	Capacity (bit)	Capacity Ratio (bit/KB)
1	13	1.02	15	1.18
2	13	1.02	15	1.17
3	14	1.11	18	1.43
4	8	0.65	9	0.73
5	6	0.49	6	0.49
6	7	0.56	9	0.72
7	32	1.74	42	2.28

Data	Even Line Shift Option		1 & 5 Pivot Line Shift Option	
	Capacity (bit)	Capacity Ratio (bit/KB)	Capacity (bit)	Capacity Ratio (bit/KB)
8	11	0.88	12	0.96
9	13	1.02	15	1.17
10	8	0.64	9	0.72
11	1,817	14.77	2,334	18.98
12	3,771	15.52	4,848	19.95
13	7,543	15.98	9,696	20.54
14	11,314	16.09	14,547	20.69

Font size experiment results can be seen in Table V. Although not directly proportional to straight, but the results to show size 8 on Sundanese Unicode font 2013 resulted in the most number of sliding rows for all the row options. While the size 20 on the Sundanese Unicode font 2013 produces the least number of shear rows.

TABLE V. CAPACITY BASED ON FONT SIZE

Font Size	Shift Option	
	Even Line	1 & 5 Pivot Line
8	3942	4731
9	3931	4587
10	3918	4410
11	3771	4848
12	3891	4488
13	3879	4848
14	3865	4215
15	3673	4407
16	3636	4848
17	3828	3828
18	3592	4041
19	3803	4278
20	3535	4545

D. Robustness Experiment

Stego text is tested for endurance in maintaining its information content against printing and scanning attacks. Robustness testing is done by extraction of stego text that has been attacked or destructed by comparing the distance of the shifted row with the control groups row. The row spacing can be measured manually by ruler as in Fig 11. The stego text is still able to withstand direct printing with the printer and reprint with a copy machine up to 2 reprints. Robustness test results as shown in Table VI.



Fig. 11. Measure the row spacing

TABLE VI. EXPERIMENT DATA

Slide Option	Functionality	Message Secret	
		Inserted	Extracted
Even Row	First Scenario	01100001	01100001
	Second Scenario	01111000	01111000
	Third Senario	00110111	00110111
1 & 5 Pivot Row	First Scenario	01100001	01100001
	Second Scenario	01111000	01111000
	Third Senario	00110111	00110111

E. Fidelity Experiment

Fidelity experiment aims to check the similarity between cover text and stego text. The PDF cover text is compared with PDF stego text as in Table VII. The 35 respondents is answering questions about the cover text and stego text similarity and calculated using Mean Opinion Score (MOS). MOS test results show the value of 3, which means the level of similarity of stego text and cover text is good enough although the size difference around 45% .

TABLE VII. EXPERIMENT DATA

Slide Option	Data	Secret Message	MOS	PDF Size (kb)			% Diff
				Cover Text	Stego Text	Δ	
Even Row	8 th data	0	3.23	16.1	8.58	7.52	46.7
	9 th data	x	3.20	16.0	8.71	7.39	45.9
1 & 5 Pivot Row	8 th data	0	3.11	16.1	8.63	7.47	46.4
	9 th data	y	3.09	16.0	8.73	7.37	45.8

F. Recovery Experiment

Recovery experiment is done by extracting the stego text to retrieve the secret message. The secret intangibility level is viewed from the secret message of the extraction process compared to the final secret message. The final final message may be different from the initial secret message because the system will add a random binary bit if the number of rows in the cover text is more than the binary bit length of the secret message. Recovery experiment will determine the level of integrity of the secret message when extracted back from the stego text. Cover text used for scenarios 1 through 6 is "cover text.docx" (12 kB) while the rest of it uses "cover text.docx" (18 kB).

TABLE VIII. EXPERIMENT DATA

Scenario	Secret Message		
	Inserted	Final	Extracted
1	x	x	x
2	di	di	di
3	ok	ok	ok
4	9	9	9
5	=	=	=
6	@	@	@
7	test	test	test
8	sony	sonyD	sonyD
9	cavs	cavs«	cavs«
10	empat	empat	empat

V. CONCLUSION

- 1) The insertion of information in Sundanese text document is done by inserting the binary bit of the secret message by changing the position of the line vertically. Line position shifting can be done by swapping Unicode blocks of Sundanese characters. The extraction process in the stego text is done by checking the Unicode blocks of rows that are considered to be shifted.
- 2) The improved Line Shift Coding has higher capacity (1.39 bits / KB) than the even shifted line options.
- 3) The level of resemblance of cover text and stego text is judged sufficiently from MOS test result which get value around 3 of scale 5.
- 4) The secret message extracted from the stego text can still be obtained intact even though it has undergone a bribery attack of printing and scanning up to 2 times with a photocopy machine.

ACKNOWLEDGMENT

This research is funded by Departement of Informatics, Faculty of Information and Communication Technology, Institut Teknologi Sepuluh Nopember (ITS), Kementerian Riset, Teknologi, dan Pendidikan Tinggi, in accordance with the Contract of Research Number: 1749/PKS/ITS/2018.

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